KAIZEN MATHS

Lesson starter

Algebraic Fractions

Simplifying Algebraic Fractions

Name _____

S1

Simplify:

$\frac{3x(2x-7)}{15x}$

S2

Simplify:

$\frac{x^2+x-12}{2(x-3)}$

S3

Simplify:

$\frac{9x(4x+3)}{12x}$

S4

Simplify:

$\frac{4x^2-20x+24}{5(x-3)}$

Show your working

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Exit ticket

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E1

Simplify:

$\frac{12x(3x-5)}{8x}$

E2

Simplify:

$\frac{3x^2-3}{2(x+1)}$

E3

Simplify:

$\frac{5x(3x+7)}{4x}$

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ALG-49C4-L1 · E

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Teacher answers & working

Algebraic Fractions · Simplifying Algebraic Fractions

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S1

Simplify:
$\frac{3x(2x-7)}{15x}$

Answer:
$\frac{(2x-7)}{5}$

Working:

Step 1: Look for a factor that multiplies the whole numerator and the whole denominator.
$\frac{3x(2x-7)}{15x}$

Step 2: The common factor is 3x. The remaining numerical factors are 1 and 5.
$\frac{3x\times(2x-7)}{3x\times5}$

Step 3: Cancel the common factor 3x from the numerator and denominator.
$\frac{3x\times(2x-7)}{3x\times5} = \frac{(2x-7)}{5}$.

Step 4: The simplified expression is $\frac{(2x-7)}{5}$.

S2

Simplify:
$\frac{x^2+x-12}{2(x-3)}$

Answer:
$\frac{(x+4)}{2}$

Working:

Step 1: Factorise the quadratic numerator before trying to cancel.

$x^2 + x - 12 = (x + 4)(x - 3)$

Step 2: Rewrite the fraction using the factorised numerator.
$\frac{x^2+x-12}{2(x-3)} = \frac{(x+4)(x-3)}{2(x-3)}$

Step 3: Cancel the common factor (x - 3).
$\frac{(x+4)(x-3)}{2(x-3)} = \frac{(x+4)}{2}$.

Step 4: The simplified expression is $\frac{(x+4)}{2}$, with x not equal to 3 because the original denominator cannot be zero.

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S3

Simplify:
$\frac{9x(4x+3)}{12x}$

Answer:

$\frac{3(4x+3)}{4}$

Working:

Step 1: Look for a factor that multiplies the whole numerator and the whole denominator.

$\frac{9x(4x+3)}{12x}$

Step 2: The common factor is 3x. The remaining numerical factors are 3 and 4.

$\frac{3x\times3(4x+3)}{3x\times4}$

Step 3: Cancel the common factor 3x from the numerator and denominator.

$\frac{3x\times3(4x+3)}{3x\times4} = \frac{3(4x+3)}{4}$.

Step 4: The simplified expression is $\frac{3(4x+3)}{4}$.

S4

Simplify:
$\frac{4x^2-20x+24}{5(x-3)}$

Answer:

$\frac{4(x-2)}{5}$

Working:

Step 1: Factorise the quadratic numerator before trying to cancel.

$4x^2 - 20x + 24 = 4(x - 2)(x - 3)$

Step 2: Rewrite the fraction using the factorised numerator.

$\frac{4x^2-20x+24}{5(x-3)} = \frac{4(x-2)(x-3)}{5(x-3)}$

Step 3: Cancel the common factor (x - 3).

$\frac{4(x-2)(x-3)}{5(x-3)} = \frac{4(x-2)}{5}$.

Step 4: The simplified expression is $\frac{4(x-2)}{5}$, with x not equal to 3 because the original denominator cannot be zero.

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E1

Simplify:
$\frac{12x(3x-5)}{8x}$

Answer:

$\frac{3(3x-5)}{2}$

Working:

Step 1: Look for a factor that multiplies the whole numerator and the whole denominator.
$\frac{12x(3x-5)}{8x}$

Step 2: The common factor is 4x. The remaining numerical factors are 3 and 2.
$\frac{4x\times3(3x-5)}{4x\times2}$

Step 3: Cancel the common factor 4x from the numerator and denominator.
$\frac{4x\times3(3x-5)}{4x\times2} = \frac{3(3x-5)}{2}$.

Step 4: The simplified expression is $\frac{3(3x-5)}{2}$.

E2

Simplify:
$\frac{3x^2-3}{2(x+1)}$

Answer:

$\frac{3(x-1)}{2}$

Working:

Step 1: Factorise the quadratic numerator before trying to cancel.

$3x^2 - 3 = 3(x - 1)(x + 1)$

Step 2: Rewrite the fraction using the factorised numerator.
$\frac{3x^2-3}{2(x+1)} = \frac{3(x-1)(x+1)}{2(x+1)}$

Step 3: Cancel the common factor (x + 1).
$\frac{3(x-1)(x+1)}{2(x+1)} = \frac{3(x-1)}{2}$.

Step 4: The simplified expression is $\frac{3(x-1)}{2}$, with x not equal to -1 because the original denominator cannot be zero.

Teacher answers & working

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E3

Simplify:
$\frac{5x(3x+7)}{4x}$

Answer:

$\frac{5(3x+7)}{4}$

Working:

Step 1: Look for a factor that multiplies the whole numerator and the whole denominator.
$\frac{5x(3x+7)}{4x}$

Step 2: The common factor is x. The remaining numerical factors are 5 and 4.
$\frac{x\times5(3x+7)}{x\times4}$

Step 3: Cancel the common factor x from the numerator and denominator.
$\frac{x\times5(3x+7)}{x\times4} = \frac{5(3x+7)}{4}$.

Step 4: The simplified expression is $\frac{5(3x+7)}{4}$.